Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	27 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Smart Lender - Loan Eligibility Prediction System

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web interface for loan application and prediction results	HTML, CSS, Bootstrap, JavaScript
2	Application Logic-1	Form handling, validation, request processing	Python, Flask
3	Application Logic-2	Data preprocessing and feature engineering	Pandas, NumPy, Scikit-Learn
4	Application Logic-3	Loan eligibility prediction engine	XGBoost, Decision Tree, Random Forest, KNN
5	Database	Store applicant details and prediction results	SQLite / MySQL
6	Cloud Database	Database service on cloud	IBM Cloud Database / MySQL
7	File Storage	Store dataset, trained model and reports	Local File System
8	External API-1	Optional credit verification service	Credit Bureau API (Optional)
9	External API-2	Optional applicant verification service	Aadhaar Verification API (Optional)
10	Machine Learning Model	Predict loan approval eligibility	XGBoost Classifier
11	Infrastructure (Server / Cloud)	Application deployment and hosting	IBM Cloud, Flask Server

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks and libraries used in the project	Flask, Scikit-Learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn
2	Security Implementations	Input validation, secure request handling, data protection	Flask Validation, HTTPS, Password Encryption
3	Scalable Architecture	Three-layer architecture separating UI, application logic and ML model	3-Tier Architecture, Flask
4	Availability	Web application accessible anytime through browser	IBM Cloud Hosting
5	Performance	Fast prediction response using trained model and optimized preprocessing	XGBoost, Pandas, NumPy

Technical Architecture Description: